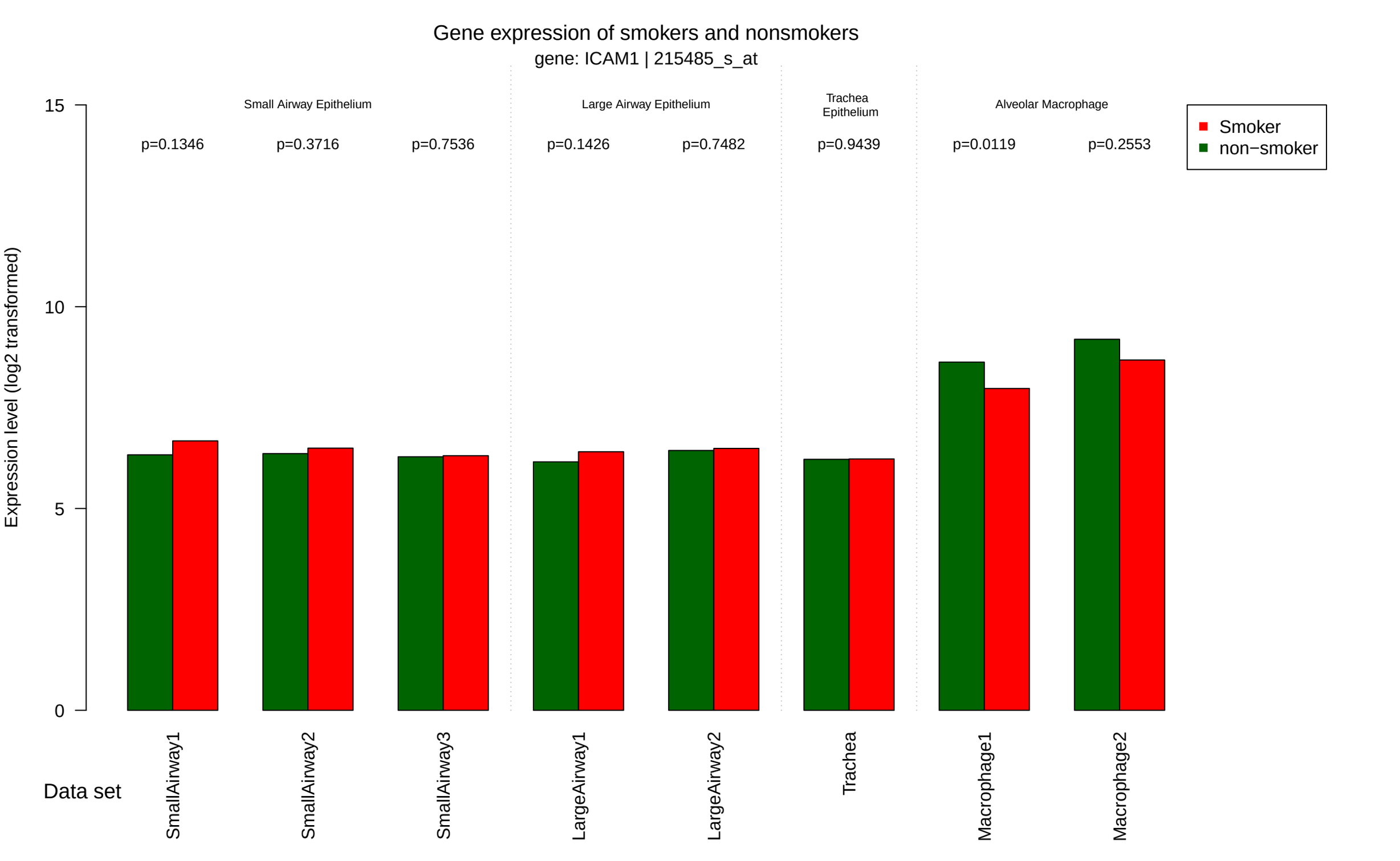
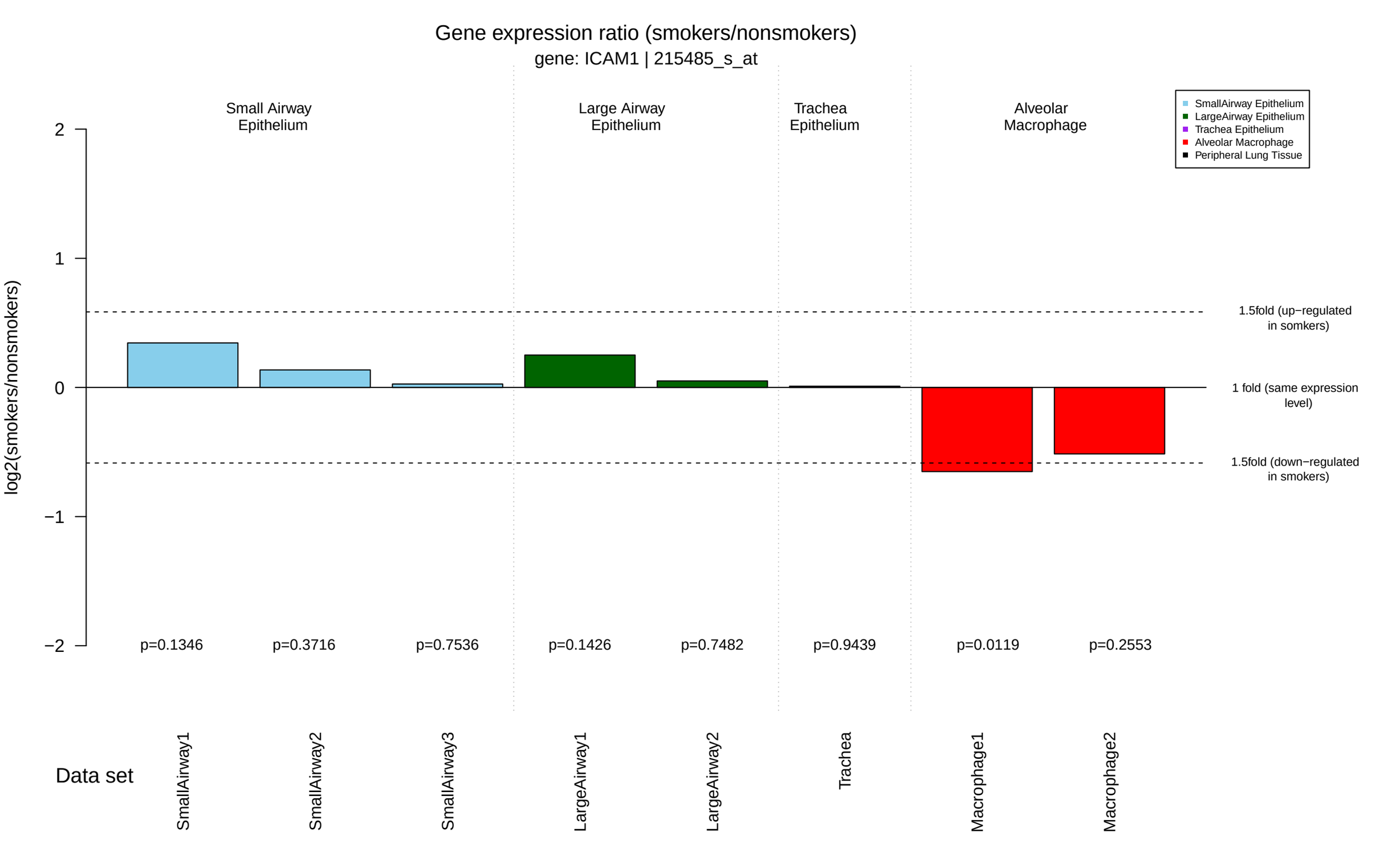
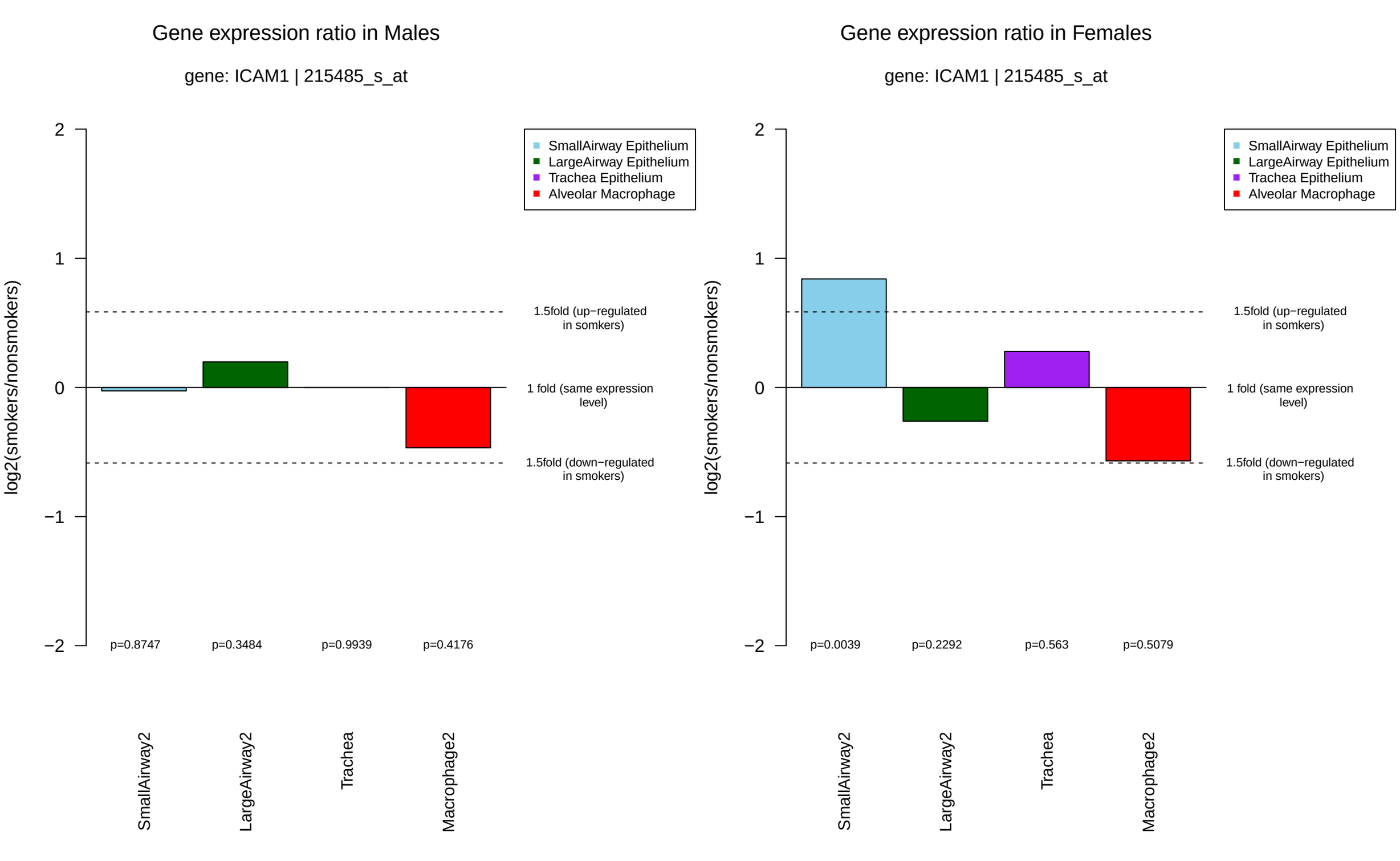
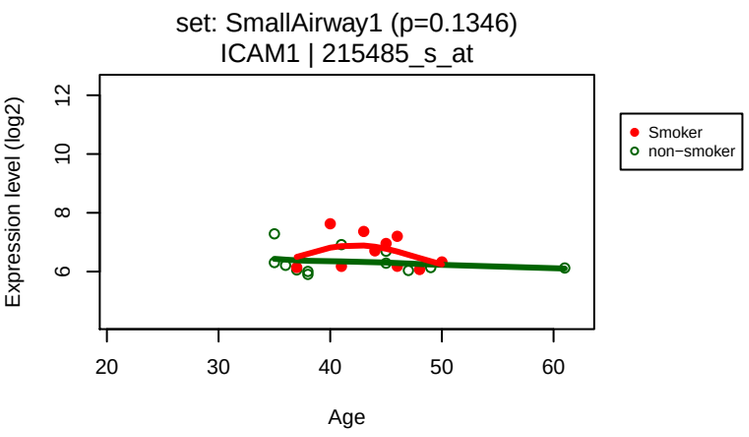
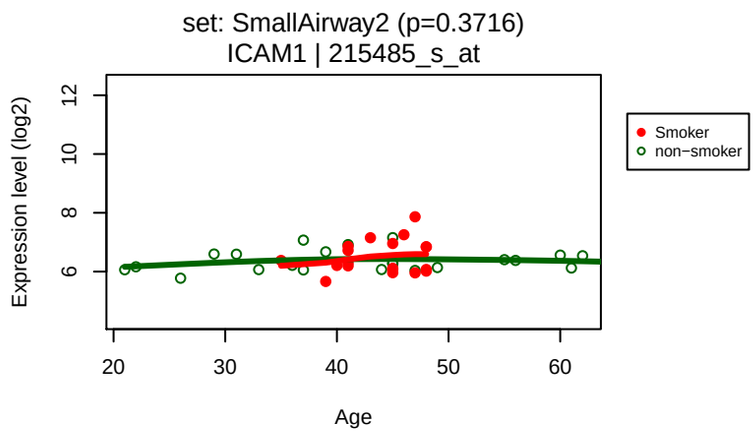
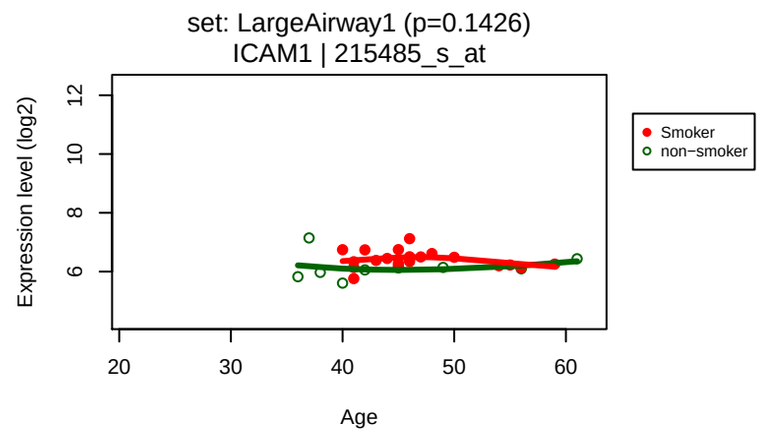

Gene expression of smokers and nonsmokers
gene: ICAM1 | 215485_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1346
p=0.3716
p=0.7536
p=0.1426
p=0.7482
p=0.9439
p=0.0119
p=0.2553
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

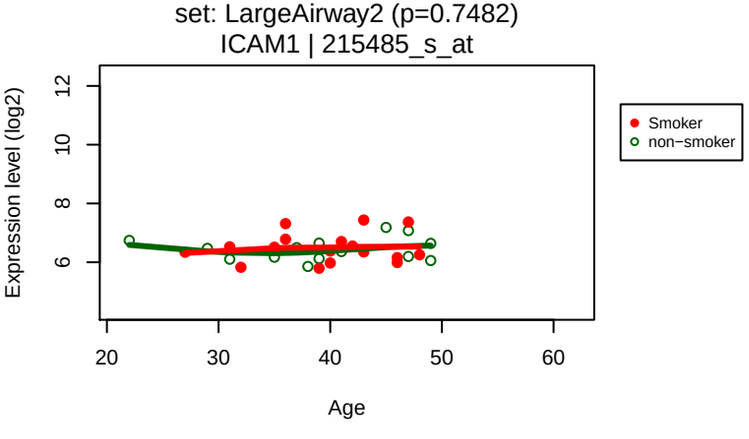

Gene expression ratio (smokers/nonsmokers)
gene: ICAM1 | 215485_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1346
p=0.3716
p=0.7536
p=0.1426
p=0.7482
p=0.9439
p=0.0119
p=0.2553
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

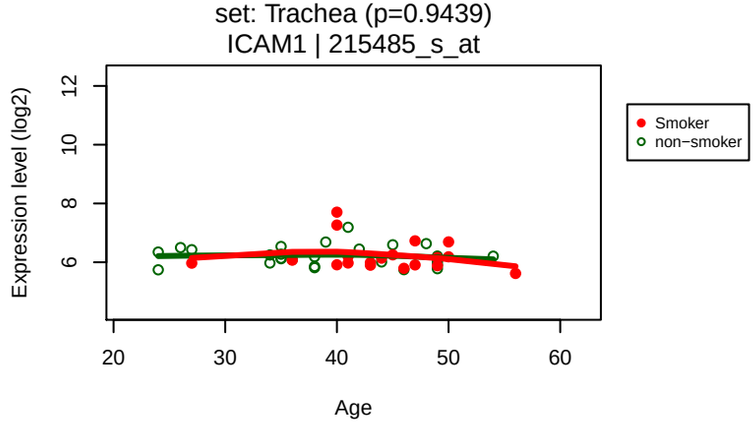

Gene expression ratio in Males
gene: ICAM1 | 215485_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8747
p=0.3484
p=0.9939
p=0.4176
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ICAM1 | 215485_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0039
p=0.2292
p=0.563
p=0.5079
SmallAirway2
LargeAirway2
Trachea
Macrophage2

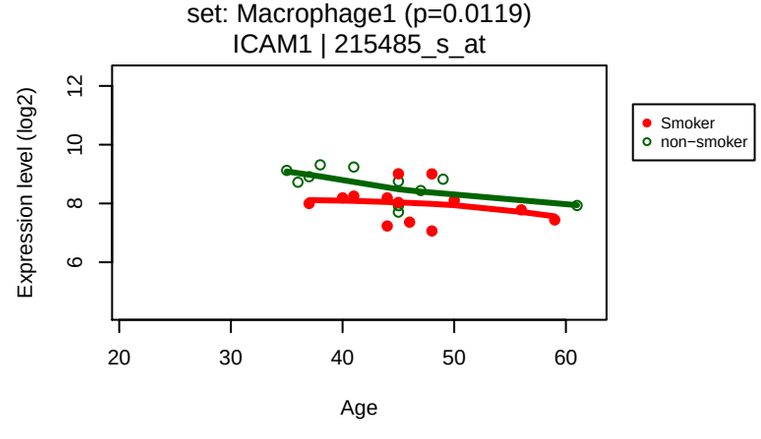

set: SmallAirway1 (p=0.1346)
ICAM1 | 215485_s_at
Expression level (log2)
Age
Smoker
non−smoker

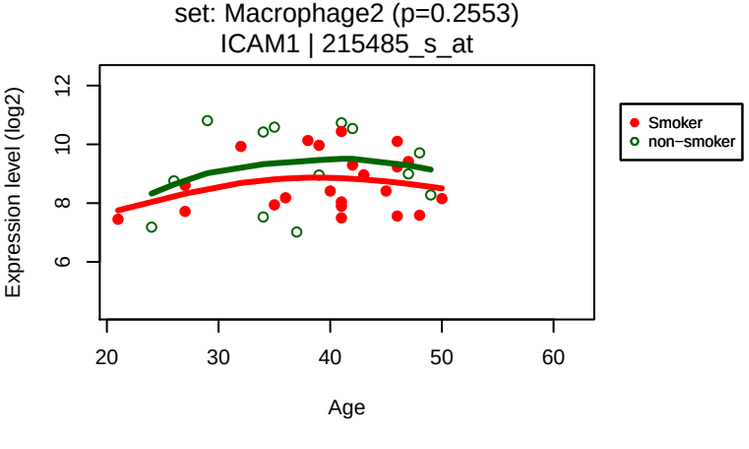

set: SmallAirway2 (p=0.3716)
ICAM1 | 215485_s_at
Expression level (log2)
Age
Smoker
non−smoker

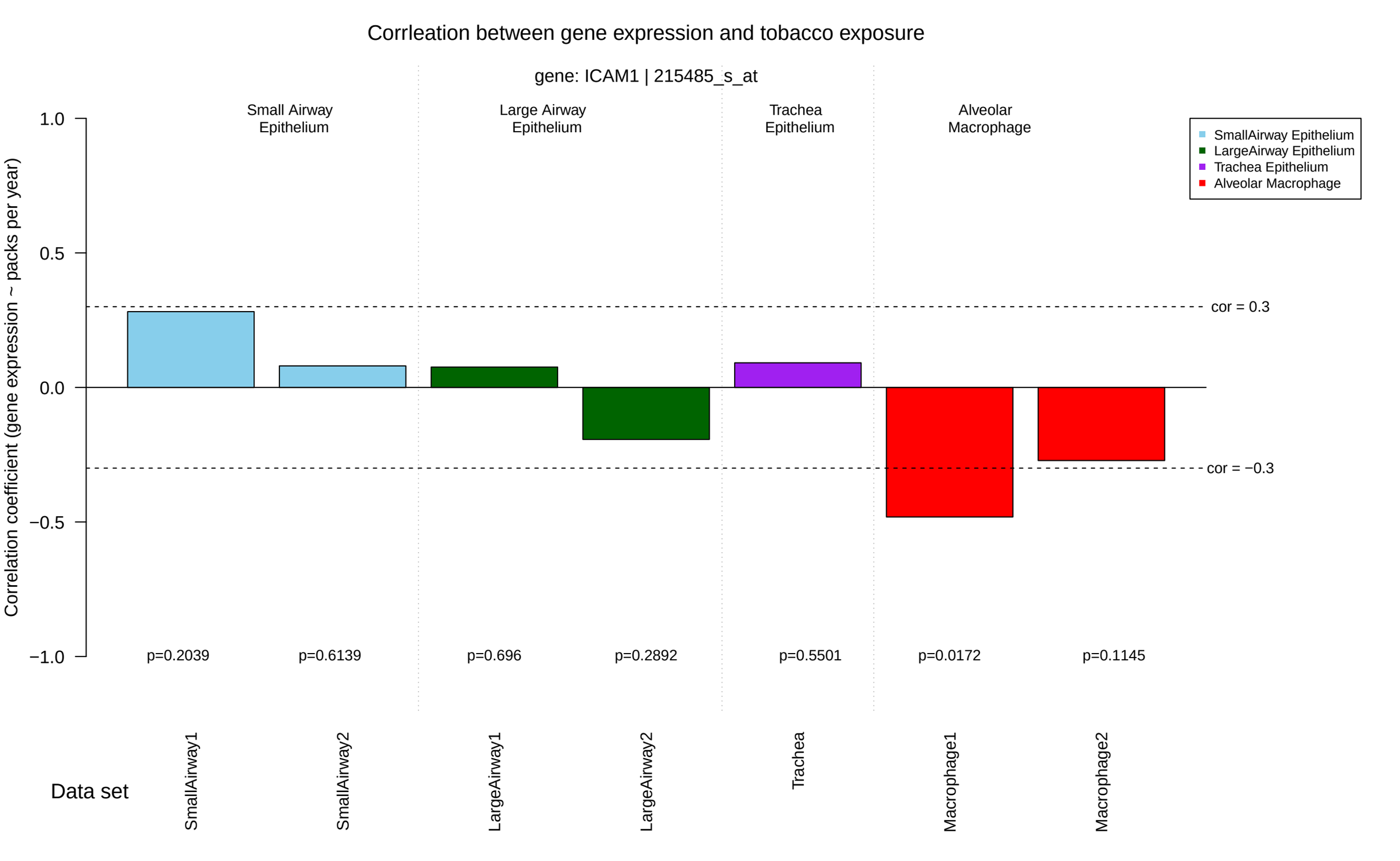

set: LargeAirway1 (p=0.1426)
ICAM1 | 215485_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7482)
ICAM1 | 215485_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9439)
ICAM1 | 215485_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0119)
ICAM1 | 215485_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2553)
ICAM1 | 215485_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ICAM1 | 215485_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2039
p=0.6139
p=0.696
p=0.2892
p=0.5501
p=0.0172
p=0.1145
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2